Table A1.1 Systematic developments in physics over centuries (*Cont*)

▪ Copernicun Revolution ▪ Heliocentric model (Not hypothesis. It provides simplest explanation than Ptolemy model for motion of stars and Planets) ▪ Accurate astronomical datas ▪ Laws of Planetary motion	15th Century AD AD 1543 (Copernicus) Tycho Brahe Kepler
▪ Law of inertia ▪ Telescope observation(Founder of modern observational astronomy) ▪ Calculation of time period of the Moons of the Jupiter ▪ Earth is not flat ▪ All object fall to Earth at the same rate (Disproved Aristotle's argument) ▪ Law of inertia(Force need not required to maintain the motion- disproved Aristotle's argument) ▪ Pendulum, inclined plane experiments ▪ Study of projectile motion ▪ Introduction of 'controlled experiments'	Gallieo Galilei (1564- 1642)
▪ Introduction of Cartesian coordinate system ▪ Idea of analytical geometry	Rane Descarte (1596-1650)
▪ 17th and 18thcentury Development	
▪ Laws of motion ▪ Quantitative idea of motion ▪ Law of gravitation ▪ Development of calculus independent of Leibnitz ▪ Founder of modern optics (reflection, dispersion, prism) ▪ Light consists of minute particles 'corpuscules' ▪ Derivation of Kepler laws ▪ Greatest book 'The principia mathematica'(1687) ▪ Wave theory of light	Isaac Newton (1642-1727) Christian Huygens; Father of modern optics

Appendix-2.indd 292 28-01-2020 14:14:38